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ORIGINAL ARTICLE



Analysis of the clinical value of serum MMP-9 and VEGF expression levels in the prenatal diagnosis of patients with aggressive placenta previa

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ABSTRACT

Objective: To investigate the diagnostic efficacy and detection value of matrix metalloproteinase-9 (MMP-9) and VEGF in menacing pernicious placenta previa (PPP).

Method: Among all the cases of PPP, a critical condition within the Placenta Accreta Spectrum (PAS) caused by aberrant implantation of the placenta in the uterine wall, which were analyzed between April 2021 and March 2023, there were sixty-three cases. The control group consisted of those sixty-three women who had a normal placenta. Serum levels of MMP-9 and VEGF were measured and compared in both groups. The expression levels of MMP-9 and VEGF were analyzed along with ultrasound scores related to different degrees of placental implantation. Comparisons between groups were performed using *t*-tests and one-way ANOVA. The diagnostic efficacy of each of the indicators was determined using receiver operating characteristic (ROC) curves by calculating the area under the curve (AUC) and Youden's index.

Results: MMP-9, VEGF expression, and ultrasound scores of pregnant women in the PPP group were significantly higher than those in the control group ($p < 0.05$). Logistic regression analysis demonstrated that MMP-9, VEGF, and ultrasound scores were significantly associated with PPP ($p < 0.05$). ROC curves indicated that serum MMP-9, VEGF, and ultrasound scores predicted the AUC of 0.802, 0.817, and 0.983 for PPP, respectively. The Youden's index values were 0.492, 0.540, and 0.826, respectively.

Conclusion: MMP-9, VEGF, and ultrasound scores help predict placental implantation in PPP, which, in turn, provides significant support for clinical understanding.

Abbreviations: AUC: area under the ROC curve; PPP: pernicious placenta previa; MMP-9: matrix metalloproteinase-9; YDI: Youden index.

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Introduction

As the standard of living improves, there is an increase in the number of macrosomic babies and a corresponding rise in the rate of obstructed labor [1,2]. Additionally, assisted reproductive technology has contributed to multiple pregnancies [3]. Pregnant women often fear labor pains and the potential for failed delivery attempts, which can lead to cesarean sections. Superstitions also play a role in influencing pregnant women and their families when choosing the timing of delivery. These factors have collectively contributed to an increase in the number of cesarean sections. A WHO survey revealed a dramatic rise in cesarean deliveries, now accounting for 47.2% of the maternal population, with one-third of these surgeries performed

without clear medical indication [4]. In China, the C-section rate has also escalated, rising from 28.8% in 2008 to 34.9% in 2014 and reaching 36.7% in 2018 [5]. This misuse of cesarean sections has emerged as a significant health issue within society.

Placenta Previa is a complication of pregnancy in which the placenta lies in the lower uterine segment after 28 weeks of gestation. This condition may lead to placental coverage of the internal cervical opening, presenting risks of large-volume hemorrhage for both mother and child. Reported incidences of placenta previa range from 0.24% to 1.57% in the United States and 0.28% to 1.5% internationally, with a marked upward trend. The most serious type is pernicious placenta previa (PPP), which occurs in subsequent pregnancies following cesarean delivery and may include

placental implantation and, in worse cases, penetration of placental villi into the uterine wall. This can lead to severe maternal complications that may include hemorrhage, shock, uterine perforation, secondary infection, and even death [6]. Both placenta previa and previous uterine scar due to the prior cesarean delivery are well-established high-risk factors for placenta implantation. It is evidenced from clinical data that the risk of placenta implantation in patients with PPP can be as high as 50% [7]. The recent rise in incidences of PPP has coincided with a general rise in the incidence of cesarean births in China, posing great challenges to clinical practice [8]. International guidelines by FIGO encourage formal terminology and a uniform pathological grading of PAS involving conditions like PPP and placenta previa, keeping terminologies in harmony while assisting toward improved diagnostic consistency and patient management. The first evidence appears in the literature from the seventeenth century, where the contributions by obstetricians such as William Smellie [9] have been very enlightening. The impending PPP is a most dangerous desolation for the safety of the mother and child in the postpartum period. The maternal morbidity and mortality in cases of PPP are largely attributed to uncontrollable bleeding, leading to severe complications of toxic shock, disseminated intravascular coagulation, multiorgan system failures, hysterectomy, intensive care unit admission, and even neonatal asphyxia and maternal and infant death [10–12]. These conditions, if allowed to persist without management, are relevant to adverse pregnancy outcomes. Various factors accompany the clinical emergence of PPP, including the comparatively short duration of its growth, inadequacies in maternal pregnancy examinations, and insufficient evidence with adequate sample sizes. In turn, this sounds a continual argument in respect to both diagnosis and treatment on national and international levels. Current clinical understanding maintains that PPP as part of PAS when combined with placental implantation might be due to the embryonic shortcomings in meconium, over-aggressive trophoblast invasion, and dysregulation in placental neovascularization [13]. Vascular endothelial growth factor (VEGF) is a cytokine that promotes mitosis in vascular endothelial cells and is produced by placental chorionic syncytiotrophoblasts; it has a significant part in the chorionic angiogenesis of the placenta. Abnormal expression of VEGF can alter the invasive capability of placenta trophoblasts by making them invasive or superficial [14]. Matrix metalloproteinase-9 (MMP-9) is a gelatinase classified as a Zn²⁺-containing metalloenzyme secreted principally by trophoblast

cells. It plays an important role in the invasion and metastasis of malignant tumor cells, and it exhibits similar behavior to tumor cell invasion [15]. This study aims to determine the relationship of serum MMP-9 and VEGF levels to severe placenta previa, especially PPP, in patients during late pregnancy, hence ascertaining the clinical significance for prenatal diagnosis. The subject study is unique in its prospect of investigating the diagnostic efficacy and detection value of serum MMP-9 and VEGF levels, specifically in PPP, a dramatically increasing entity with severe complications during late pregnancy. Unlike most previous studies focused on placental abnormalities in general or relying on limited diagnostic criteria, our study integrates serum biomarkers with ultrasound scoring, aiming for efficient diagnosis. Our study confirms elevated expression levels of MMP-9 and VEGF, having investigated a relatively large study population of 63 patients with PPP and an equal-sized control group that establishes predictive value by Receiver operating characteristic (ROC) curve analysis. To our knowledge, this integrative method meets international PAS guidelines. It offers a new multi-faceted avenue for diagnosis that may result in increased prenatal care for and improved management of women at risk for developing PPP, thus addressing a huge gap in existing clinical research.

Materials and methods

General information about patients

The PPP group comprised 63 pregnant women clinically diagnosed with PPP, admitted from January 2021 to October 2023. The study was approved by the Ethics Committee of the First People's Hospital of Tianmen City, and all participants provided written informed consent.

Inclusion and exclusion criteria

The PPP group consisted of patients pathologically confirmed to have a disorder of the Placenta Accreta Spectrum (PAS). It is characterized by a history of cesarean section, placenta previa with the current pregnancy, and attachment of the placenta to the uterine incision scar with/without placental implantation. Postoperative pathology confirmed the presence of chorionic tissue in the myometrium. The control group consisted of pregnant women with normal placentas. All singleton pregnancies with complete clinical data were included. All participants were informed of the study protocol and gave written informed consent voluntarily. Exclusion criteria: (1) Combined with

hypertensive disease in pregnancy, gestational diabetes mellitus, and intrahepatic cholestasis in pregnancy and other pregnancy comorbidities; (2) Twin pregnancies; (3) Accompanied by malignant neoplastic disease; (4) Existence of hematologic diseases; (5) Accompanied by severe liver and kidney diseases before pregnancy; (6) Combined with anemia.

Method

All examinations were conducted using a GE E8 color Doppler ultrasound machine. Trained obstetric sonographers performed the ultrasound assessments following standardized operating procedures to ensure reliability and consistency. The transabdominal probe operated in the frequency range of 3.0–5.0 MHz, and the transvaginal probe at 8.0–10.0 MHz. These settings allowed assessment of the following important parameters: position of the placenta, placenta thickness, posterior hypoechoic zone of the placenta, the lining of the bladder, placenta's fossa, blood flow signal from the placenta's fundus, cervical sinus, and cervix shape.

Interreader Variability: It adds robustness and provides evidence of reproducibility for the ultrasound score through the assessment of agreement among independent sonographers.

Color ultrasound diagnostic criteria included the following: The muscular layer of the anterior uterine wall scar was extremely thin at the site of placenta attachment. Areas of thickening were round, either big or small, with varieties in shapes appearing vortex-like and abundant with blood flow. The retro-placental area was partially or completely absent, with a profusion of blood flow signals. The vessels between the placenta and chorionic villi appeared tortuous and disordered in lateral views. Furthermore, strong echoes were partially interrupted between the uterus and the bladder.

The pathologic criteria for diagnosing placenta previa with implantation were set in opposition to the gold standard based on the results of postoperative pathology. The tortuous, full, and dilated lower uterine vessels were found during surgery. The placenta could not extrude spontaneously from the uterine wall, which caused significant hemorrhage. The surface of the abruption of the placenta was rough, with lobules missing, while pathological examination confirmed chorionic tissue in the myometrium.

Detection of MMP-9 and VEGF Levels: All fasting venous blood samples were collected in the morning from all participants. The blood samples were centrifuged for 10 min at 3,000 rpm and stored at -70°C until further analysis. For the measurement of MMP-9

and VEGF, ELISA kits were used according to the manufacturer's protocol. This method ensured accurate and reliable measurements of these particular biomarkers.

Observation indicators

This section discusses several key study aspects of serum MMP-9 and VEGF expression levels in pregnant women. Firstly, it examines the overall expression levels of serum MMP-9 and VEGF. The study then delves into the expression levels of these markers, specifically in pregnant women experiencing different degrees of placental implantation. The assessment of placental implantation was conducted using ultrasound scores derived from the "Ultrasound Scoring Scale for Placental Implantation" (Patent Application No. 2016107944516), developed by the Third Hospital of Peking University. Furthermore, the study explores the correlation between the serum levels of MMP-9 and VEGF and the ultrasound scores. Lastly, it evaluates the predictive value of serum MMP-9, VEGF, and color ultrasound in determining the degree of placental implantation.

Statistical methods

The data were statistically analyzed using SPSS 27.0 software. Descriptive statistics were expressed as mean $\pm$ standard deviation ($\pm$ s). Comparisons between two groups were performed using the independent *t*-test, and one-way ANOVA was utilized for comparisons among multiple groups, followed by the LSD *post hoc* test. The correlation analysis and model building were based on logistic regression. The area under the ROC curve (AUC) realized the model prediction efficiency evaluation index. In the case of $\text{AUC} > 0.5$, the closer the AUC is to 1, the better the diagnostic effect of the model is; the fluctuation of AUC between 0.5 and 0.7 suggests lower accuracy, the fluctuation of AUC between 0.7 and 0.9 suggests some accuracy, and the model has higher accuracy when the AUC is above 0.9. At the same time, the maximum value of the Youden index (YDI) was used as the cutoff value, and the Delong test was used to evaluate the area under different ROC curves—test level $\alpha=0.05$.

Results

Baseline information of the patients

No statistically significant differences were observed from the baseline characteristics (age, gestational age, number of pregnancies, number of deliveries)

among patients of the PPP and control groups (Table 1, all $p > 0.05$).

Table 1. Results of baseline information.

Indicators	PPP (n=63)	Control (n=63)	t	P
Age (year)	33.09±4.84	31.65±4.61	1.714	0.089
Gestational weeks	37.38±2.34	37.65±2.17	0.671	0.503
Number of pregnancies	3.73±1.18	3.52±1.11	1.013	0.313
Number of births	1.65±0.68	1.59±0.64	0.542	0.589

Abbreviation: PPP: pernicious placenta previa.

Table 2. Results of MMP-9, VEGF expression and ultrasound score.

Group	MMP9 (pg/mL)	VEGF (pg/mL)	Ultrasound score
PPP (n=63)	412.40±119.19	287.85±91.32	7.59±3.66
Control (n=63)	276.58±102.27	189.77±60.21	1.29±0.63
F	6.864	7.117	13.444
P	<0.001	<0.001	<0.001

Abbreviations: MMP9: Matrix Metalloproteinase 9; PPP: pernicious placenta previa; VEGF: Vascular Endothelial Growth Factor Receptor.

MMP-9 and VEGF expression and ultrasound score

Serum MMP-9 and VEGF levels were higher in women with PPP than in control pregnant women, with statistically significant differences (Table 2, all $p < 0.05$). This is again emphasized in Figure 1, which shows higher levels of MMP-9 and VEGF in the PPP group than in the control group.

Indicator results for different degrees of placental implantation

The serum MMP-9 and VEGF levels of pregnant women with placenta accreta were higher than those of pregnant women with placenta increta. Additionally, the serum MMP-9 and VEGF levels of pregnant women with placenta increta were higher than those of pregnant women with placenta percreta differing significantly (Table 3, all $p < 0.05$).

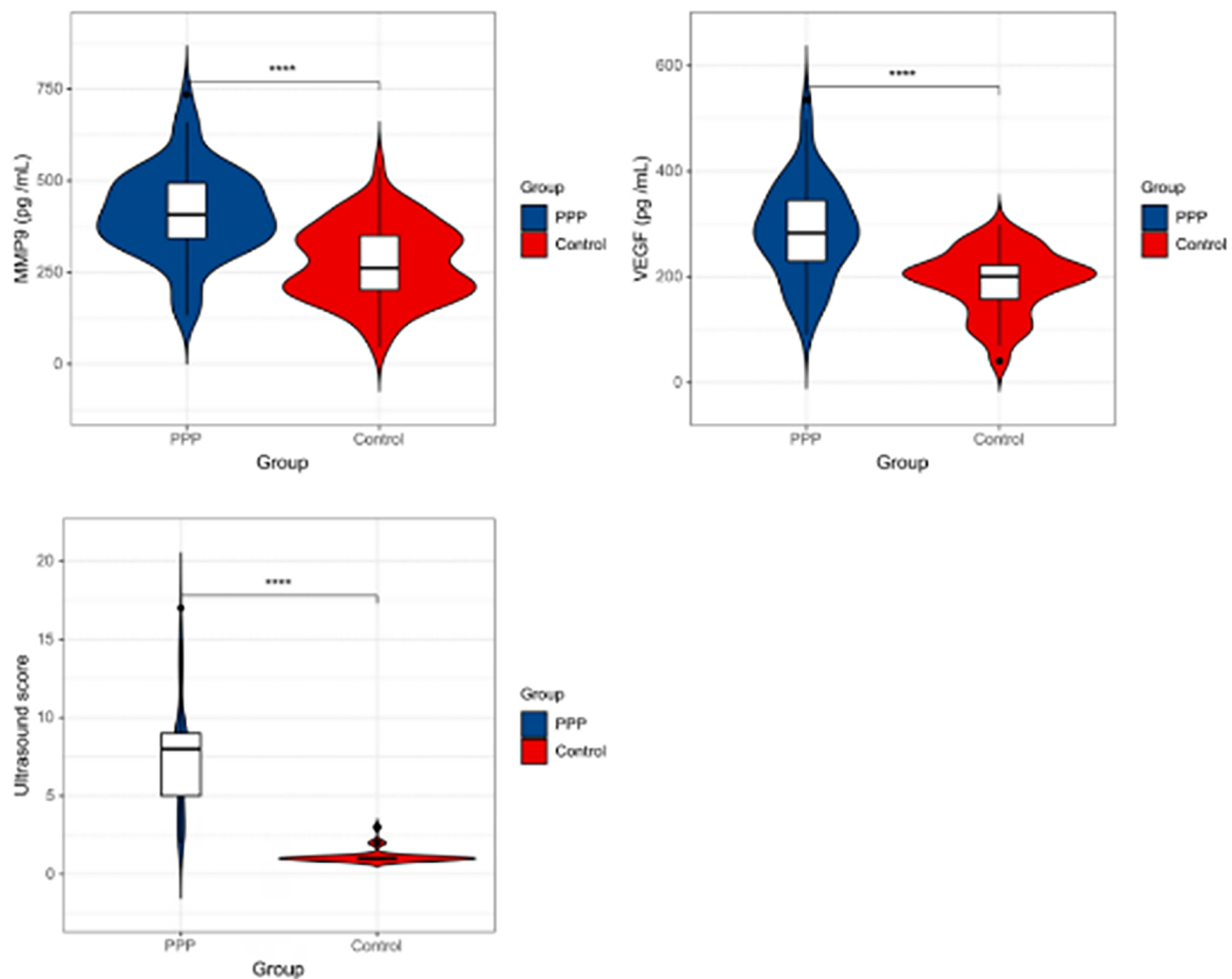


Figure 1. Comparison of MMP-9, VEGF expression, and ultrasound score.

Comparison between the MMP-9 and VEGF levels and the ultrasound score in the PPP and control groups shows statistically significant differences ($P < 0.05$)—overview of the ultrasound scoring system used to assess placental implantation.

Logistic regression analysis

Logistic regression analysis was performed with PPP as the dependent variable (pernicious placenta previa = 0, normal placenta previa = 1) and MMP-9, VEGF, and ultrasound score as independent variables. The results show that MMP-9, VEGF, and ultrasound scores can have some diagnostic values in PPP (Table 4, all $p < 0.05$).

In this study, normal placenta previa designates a condition in which the placenta is properly located at the lower segment of the uterus with no evidence of abnormal invasion or penetration into the uterus wall. The results show that MMP-9, VEGF, and ultrasound scores have specific diagnostic values in PPP (Table 4, all $p < 0.05$).

MMP-9, VEGF, and color ultrasound diagnostic results

At the end of the study, the sensitivities were 65.1%, 79.4%, and 92.1%, and the specificities were 84.1%, 74.6%, and 90.5%, respectively, with the Youden index values being 0.492, 0.540, and 0.826. The AUCs were all found to be greater than 0.8 and closer to 1, which indicates that the model is highly accurate (Table 5, Figure 2).

Table 3. Comparison of MMP-9, VEGP and ultrasound scores in pregnant women with different degrees of placental implantation.

Group	MMP9 (pg/mL)	VEGF (pg/mL)	Ultrasound score
Placenta penetration (n = 11)	534.62 ± 122.76	376.24 ± 84.34	13.18 ± 1.60
Placental implantation (n = 34)	413.45 ± 80.22	293.76 ± 77.58	8.09 ± 1.68
Placental Adhesion (n = 18)	335.72 ± 120.66	222.68 ± 70.17	3.22 ± 1.06
F	13.276	13.884	282.244
P	<0.001	<0.001	<0.001

Abbreviations: MMP9: Matrix Metalloproteinase 9; VEGF: Vascular Endothelial Growth Factor Receptor.

Table 4. Logistic analysis of serum MMP-9, VEGF and color ultrasound.

Indicators	B	SE	Wald χ^2	P	OR	95%CI
MMP9	0.012	0.006	3.928	0.047	0.988	0.977–1.000
VEGF	0.024	0.012	3.877	<0.049	0.976	0.953–1.000
Color ultrasound	2.483	0.658	14.262	<0.001	0.083	0.023–0.303

Abbreviations: CI: Confidence interval; MMP9: Matrix Metalloproteinase 9; OR: odds ratio; SE: Standard error; VEGF: Vascular Endothelial Growth Factor Receptor.

Table 5. ROC Results for MMP-9, VEGF, and color ultrasound.

Indicators	AUC	SE	P	Sensitivity	Specificity	YI	95%CI
MMP9	0.802	0.039	<0.001	0.651	0.841	0.492	0.725–0.879
VEGF	0.817	0.039	<0.001	0.794	0.746	0.540	0.741–0.894
Color ultrasound	0.983	0.008	<0.001	0.921	0.905	0.826	0.968–0.998

Abbreviations: AUC: area under the receiver operating characteristic curve; CI: Confidence interval; MMP9: Matrix Metalloproteinase 9; SE: Standard error; VEGF: Vascular Endothelial Growth Factor Receptor; YI: Youden index.

Discussion

The concept of placenta previa was first introduced in 1993 by Chattopadhyay et al. It is often combined with placental implantation. Implanted placenta previa is a common cause of obstetric hemorrhage and often leads to fatal postpartum hemorrhage [16]. With the adjustment of China's fertility policy, the amount of re-pregnancy in scarred uteruses is increasing, and the incidence of PPP is gradually increasing [17]. According to relevant studies, as the number of cesarean sections increases, the incidence of placenta previa increases, and the risk of implantation of placenta previa and adverse perinatal outcomes increases [18,19]. The management of PPP is more difficult, and the prognosis of mothers in hospitals with multidisciplinary teams is significantly better. Therefore, delivering the baby to a higher-level hospital with better treatment capability is recommended. Over the years, many scholars at home and abroad have researched and investigated the risk factors of placenta previa from different levels, which are advanced pregnancy age, multiple pregnancies, and multiple abortions [20]. However, its exact pathogenesis is still unknown and may be related to endometrial lesions or injuries and poor healing of uterine scars after cesarean section [21].

In this study, a "normal placenta" is defined as one that is properly positioned within the uterus without any signs of previa or abnormal implantation. This definition serves as a baseline for comparing the pathological changes observed in cases of placenta previa.

Magnetic resonance can be used for prenatal diagnosis of PPP, but its use in clinical practice is relatively short [22]. Magnetic resonance requires a higher operator experience, is less sensitive in diagnosing adhesions, and is more expensive and time-consuming. In contrast, color ultrasound is inexpensive, easy to operate, and requires fewer technicians [23]. Many studies have shown no significant difference in the diagnostic accuracy between the two in diagnosing aggressive placenta previa [24]. Therefore, color ultrasound is easier to promote and apply in primary hospitals and is more practical in clinical practice. It is currently the primary examination method for diagnosing implantation of placenta previa [25]. The differences in the results of the clinical studies may be due to the different precision of the testing instruments and the experience of

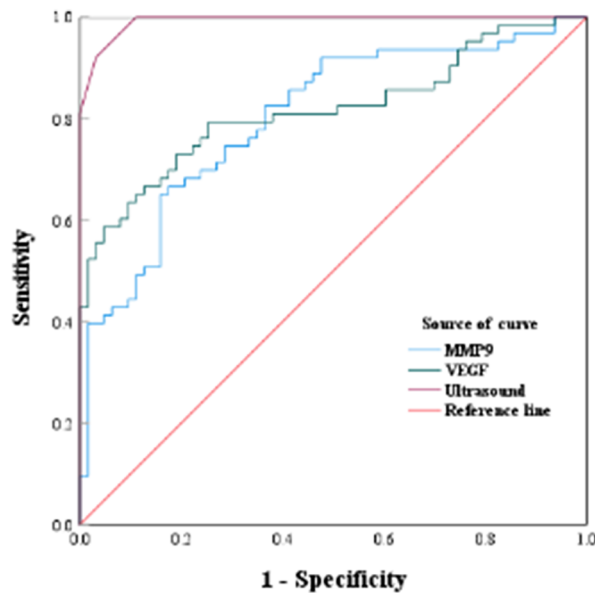


Figure 2. ROC curves of MMP-9, VEGF, and color ultrasound. ROC curves show the diagnostic value of MMP-9, VEGF, and color ultrasound for the PPP. From the AUCs, the relative accuracy of each test can be determined according to the colors: color ultrasound appears to be the most accurate.

the operators in each hospital. Meanwhile, due to the defects of color ultrasound, such as the susceptibility of color ultrasound to artifacts caused by the fetus and its appendages, the location of the placenta, and other objective factors, there are differences in the description of the type of implantation in question, which makes it easy to misdiagnose, and this is one of the reasons for the varied diagnostic results reported by the clinics. In this study, ultrasound scores were higher in the PPP group than in the control group. The ultrasound score showed an increasing trend as the menacing placenta previa implantation was higher.

Moreover, this study conforms to the recommendations published by the FIGO, International Federation of Gynecology and Obstetrics, regarding PAS. These guidelines stress the importance of standardization in diagnosing and managing PAS using uniformly recognizable terms, classifications, and diagnostic techniques, including ultrasound and pathology. This interdisciplinary assessment, performed with an ultrasound scoring system and qualitative measurement of biomarkers, including MMP-9 and VEGF, reflects global recommendations and highlights the utility of imaging in conjunction with serum markers in improving diagnostic accuracy [26]. Placenta previa is an obstetric disorder with uncertain etiology that leads to late pregnancy complications. Several possible mechanisms have been advanced to explain this condition, including mechanical factors that refer to damage of the endometrium, destruction of the basal lamina or

scarring, and biological factors that can consist of abnormally maternal responses during trophoblastic growth [27]. During implantation and development of the infant, the embryonic trophoblast invades the small uterine spiral arteries and vascular lumen, transferring retrogradely with the vascular endothelial cells, infiltrating the decidual membrane and myometrium, reserving or replacing the vascular endothelial cells and establishing the circulatory relationship between the mother and the fetus [28]. MMP-9 and VEGF are involved in trophoblast invasion and placental vascularization processes [29]. Cellular sources of MMP-9 in the human body include placental villous trophoblasts, neutrophils, monocytes, endothelial cells, fibroblasts, and tumor cells. MMP-9 is secreted as a precursor or zymogen and activated into final forms of expression for its biological functions [30]. MMP-9 is implicated in extracellular matrix degradation and has a role in cell adhesion, inflammatory response, trauma repair, angiogenesis, and tissue model remodeling during the development of embryonic tissues [31]. The breakdown of the endometrial extracellular matrix by MMP-9 is a crucial step in the trophoblast invasion. It may interfere with the synthesis, secretion, and degradation of MMP-9, paving the way for many diseases. Xu has also stated that MMP-9 started being secreted between the 7th and 11th weeks of gestation, thus playing an essential role in trophoblastic invasion [32, 33]. In this study, we examined the levels of MMP-9 expressed in sera from patients with placenta previa. Sera from patients with PPP had a highly significant increase in expression of MMP-9 relative to healthy controls ($p < 0.05$). Due to several advantages of serum VEGF measurements over urine samples, we chose serum over urine VEGF. Serum VEGF levels are more reliable in reflecting placental function than urine measurements, which may be influenced by hydration status and renal function, as illustrated by earlier studies, including Munoz et al. which show that serum VEGF can serve as a placenta disease biomarker by making a significant contribution to clinical judgments [34]. The MMP-9 level increased with the severity of the disease, and with more aggressiveness of placenta previa implantation, the MMP-9 level increased progressively. Nonetheless, the poor sensitivity of MMP-9 toward diagnosing PPP may be attributable to the small sample size utilized in this study. Further investigations are warranted with a larger sample size. VEGF is a dimeric glycoprotein composed of two homologous polypeptide chains. The biological activity of VEGF is mediated through receptor binding on vascular endothelial cells (VEGFR-1, VEGFR-2, and VEGFR-3). The functions of VEGF include acting as a promoter of angiogenesis, a

maintainer of vascular wall permeability and integrity, a regulator of the functions of vascular endothelial cells, and an important player in placental angiogenesis. Multiple tissues and organs of the human body having high perfusion, including the heart, spleen, kidney, and lungs-express VEGF at a fairly high level [35]. Tseng's study confirmed that VEGF receptors are present on trophoblast cells; therefore, VEGF is of great importance in trophoblast cell invasion and differentiation, as well as in the growth and development of the placenta [36]. Szewczyk's study showed that VEGF can mediate the construction of the placental circulation by regulating the erosive capacity of trophoblast cells [37]. In this study, VEGF expression was significantly higher in the PPP group than in the control group. The increase in information obtained concerning the diagnosis of PPP related to the increase in the stage of implantation of the placenta previa tended toward an increase in the level of VEGF. The ultrasound scoring system used in this study assigns scores according to the degree of severity of placental previa. The higher the score, the worse the disease progression is, which can be correlated with the clinical outcomes and inform management decisions in affected patients. This scoring system, therefore, has significant implications for the prognosis of patients with placenta previa. To recap, both MMP-9 and VEGF were upregulated in the serum of pregnant women, establishing supposed clinical value for the diagnosis of PPP. However, conditions subjected to qualification in this study's findings must wait until further clarifications are achieved. Apart from this, a couple of limitations have emerged in such studies. First, though sufficient, the sample size resulted from a single center, making it difficult for a generalized conclusion to be made. Secondly, whereas effective, aspects of placental insertion that could not be listened to were serologically based and ultrasonically assessed. Compounding these issues, data were taken retrospectively; thus, there is a diminishing ability to gauge biomarker level changes and clinical outcome changes over time without follow-up. All of these issues point to further studies with larger cohort studies and a multicenter and longitudinal follow-up of patients in future studies to support our findings further.

Conclusion

In conclusion, MMP-9 and VEGF were highly expressed in the serum of pregnant women with aggressive placenta previa, a condition characterized by abnormal placental implantation. Placental pathology, particularly

the Placenta Accreta Spectrum, encompasses a range of abnormalities where the placenta invades the uterine wall more deeply than normal, which may lead to significant clinical complications. Serum MMP-9, VEGF, and ultrasound scores effectively predict placenta implantation and can support clinical data in managing these conditions.

Acknowledgment

Not applicable.

Consent for publication

Not applicable.

Ethics approval and consent to participate

The First People's Hospital of Tianmen City approved the study. Written informed consent was obtained from all individuals included in this study.

Disclosure statement

No potential competing interest was reported by the authors.

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Availability of data and material

All data generated or analyzed during this study are included. Further inquiries can be directed to the corresponding author.

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